

1. TITLE & METADATA

- **Report Title:** Real-Time Weather Analysis
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- **Date:** September 23, 2026
- **Version Number:** 5.0

2. EXECUTIVE SUMMARY

An evaluation of the 5-city Power BI weather dashboard (Bengaluru, Hyderabad, Latur, Mumbai, and Pune) was conducted using visual diagnostics and underlying trace logs. Atmospheric safety across all monitored regions remains optimal, with Air Quality Index (AQI) values falling strictly within the "Good" category (ranging from 21 in Pune to 44 in Latur). However, high precipitation probabilities exceeding 80% and elevated UV indices above 8.0 across specific forecast days present clear operational exposure for field teams. Combining single-metric scalar card containers into unified visual blocks is recommended to reduce visual async rendering overhead and optimize dashboard performance.

3. OBJECTIVE / BUSINESS PROBLEM

- **Primary Purpose:** Deliver a detailed, multi-city analytical overview of weather patterns, rain probabilities, and air quality metrics while auditing the underlying DAX architecture and visual performance model.
- **Business Decision:** Support operational risk management for regional logistics, standardize reporting metrics, and optimize Power BI dashboard load times across regional nodes.

4. DATA SOURCES & METHODOLOGY

- **Data Source:** Live/API-driven Power BI dataset (Locations, Current, Forecast_Day, and _Measures tables) paired with Performance Analyzer execution trace logs.
- **Time Period Covered:** September 8 baseline snapshot with a 7-day extended forecast (Tuesday through Monday).
- **Sample Size:** 5 Cities (Bengaluru, Hyderabad, Latur, Mumbai, Pune) across 7 forecast days and 10 visual container components per page.
- **Tools Used:** Power BI Desktop/Service, DAX Studio, Performance Analyzer JSON analysis.
- **Cleaning Steps & Assumptions:** Context filtering applied via TREATAS() slicers, temperature metrics normalized to Celsius (°C), and trace logs filtered to isolate semantic engine execution.

5. KEY FINDINGS / ANALYSIS

- **Bengaluru Regional Summary:**

- **Current Conditions:** Ambient temperature recorded at 27.7 °C under Sunny skies, with 51% humidity, 13 Kph wind speed, and an elevated UV Index of 9.20.
- **Air Quality Profile:** Overall AQI is classified as "Good" at 32, with PM10 at 32, O3 at 64, SO2 at 3.90, PM2.5 at 14.80, CO at 264, and NO2 at 4.70.
- **Precipitation Outlook:** Rain chances range between 43% (Thursday/Sunday) and a peak of 84% on Wednesday.

- **Hyderabad Regional Summary:**

- **Current Conditions:** Ambient temperature recorded at 26.0 °C with Patchy Rain Nearby, 63% humidity, 18 Kph wind speed, and a UV Index of 6.60.
- **Air Quality Profile:** Overall AQI is classified as "Good" at 31, with PM10 at 31, O3 at 108, SO2 at 7.10, PM2.5 at 19.00, CO at 246, and NO2 at 3.90.
- **Precipitation Outlook:** High rain probability throughout the forecast, ranging from 43% (Tuesday/Sunday) up to 88% on Wednesday.

- **Latur Regional Summary:**

- **Current Conditions:** Ambient temperature recorded at 26.3 °C under Sunny skies, 50% humidity, 14 Kph wind speed, and a high UV Index of 8.30.
- **Air Quality Profile:** Overall AQI is classified as "Good" at 44 (highest among all five cities), with PM10 at 44, O3 at 69, SO2 at 4.90, PM2.5 at 17.50, CO at 133, and NO2 at 1.70.
- **Precipitation Outlook:** Rain chances range from 42% (Sunday) to a maximum of 84% on Wednesday and Saturday.

- **Mumbai Regional Summary:**

- **Current Conditions:** Ambient temperature recorded at 28.1 °C (highest observed) with Light Rain Showers, high humidity at 72%, wind speed at 7 Kph, and UV Index at 7.10.
- **Air Quality Profile:** Overall AQI is classified as "Good" at 32, with PM10 at 32, O3 at 49, SO2 at 13.00, PM2.5 at 14.00, CO at 132, and NO2 at 5.90.
- **Precipitation Outlook:** Elevated rain probability across all 7 days, peaking at 87% on Monday.

- **Pune Regional Summary:**

- **Current Conditions:** Ambient temperature recorded at 23.4 °C (lowest observed) with Patchy Rain Nearby, 68% humidity, 12 Kph wind speed, and UV Index at 7.10.
- **Air Quality Profile:** Overall AQI is classified as "Good" at 21 (lowest/cleanest air quality score), with PM10 at 21, O3 at 66, SO2 at 6.60, PM2.5 at 10.00, CO at 136, and NO2 at 3.40.
- **Precipitation Outlook:** Significant precipitation likelihood throughout the week, peaking at 88% on Monday.

6. INSIGHTS

- **Geographic & Microclimate Variance:** Coastal proximity heavily influences humidity and current temperature (Mumbai leading at 28.1 °C / 72% humidity), while higher elevated inland areas show cooler profiles (Pune at 23.4 °C).
- **Solar Exposure vs. Temperature:** High UV indices in Bengaluru (9.20) and Latur (8.30) indicate that solar radiation risk does not directly correlate with ambient temperature, requiring independent safety tracking.
- **Dashboard Execution Efficiency:** DAX engine execution times are extremely fast (<25 ms per query), but front-end container rendering overhead is expanded due to hosting scalar measures in individual shape objects.

7. RECOMMENDATIONS

- **Quick Win:** Consolidate scalar visual objects (Humidity, Wind Speed, Visibility, Pressure, UV Index, Precipitation) into modern Multi-row Cards or New Card Visuals to reduce container rendering requests.
- **Quick Win:** Add conditional formatting rules to the "Chances Of Rain" visuals to highlight days exceeding 80% probability.
- **Long-term Strategy:** Automate data refreshes via Power BI Service using API connectors to replace manual dataset updates.

8. LIMITATIONS / CAVEATS

- **Predictive Horizon:** Weather metrics are constrained to a static 7-day forecast window.
- **Snapshot Context:** Analysis represents a specific point-in-time refresh (September 8 baseline).
- **Execution Log Scope:** Performance trace files provided focus on single-location filtering events ("Pune").

9. APPENDIX

- **Dashboard DAX Architecture & Representative Examples:**

- **Scalar Current Metric Evaluation:** Curr_temp_c = CALCULATE(SELECTEDVALUE('Current'[temp_c])) — Used as the primary logic pattern across current atmospheric scalar metrics including Humidity, Wind Speed, Visibility, Atmospheric Pressure, UV Index, and Precipitation.
- **Aggregated Forecast Metric Evaluation:** For_temp_c = SUM('Forecast_Day'[forecast.forecastday.day.avgtemp_c]) — Representative aggregation pattern used to compute daily values across the 7-day forecast trend.
- **Minimum Timestamp / Text Selection:** Sunrise_date = MIN('Forecast_Day'[Sunrise_date]) — Representative single-value evaluation pattern used for fetching textual and timestamp fields like Sunrise, Sunset, Location Name (MIN('Locations'[location.name])), and Weather Condition Text (MIN('Current'[current.condition.text])).
- **Context Slicing Filter:** TREATAS({"Pune"}, 'Locations'[location.name]) — Core DAX filter expression dynamically applied across visual containers to constrain execution context to the active city.
- **Pollutant & AQI Metric Categorization:** Evaluates concentration values for PM10, PM2.5, O3, NO2, SO2, and CO to dynamically establish the overall "Good" AQI status indicator.
- **Rain Probability Percentage Distribution:** Extracts daily rain probability percentages ('Forecast_Day'[daily_chance_of_rain]) across the 7-day forecast horizon.